CCCA Cell-State Analysis — Rendeiro et al. 2020, Hematologic

Chronic Lymphocytic Leukemia · 10x · 18,452 malignant cells (7 samples)

GO / NO-GO GATES

Gate 1 — mixing & continuity — GO

Harmony(theta=10) integrates all 7 samples: same-sample neighbour fraction 0.97 -> 0.22, iLISI median 1.00 -> 4.95 of 7. 205 cells in 3 detached islands removed; one connected component. Read on the regenerated embedding + per-neighbourhood iLISI, not a scalar: samples intermix AND the manifold is continuous.

Gate 2 — module geography — GO

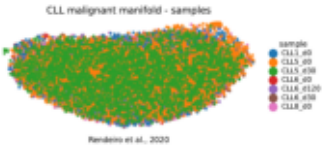
Non-branching case: a single (Gast)->Crest developmental gradient. Gast opposes Crest in space (smoothed r=-0.75; pSom weakly opposes both). Along pseudotime Gast falls (rho -0.64) while Crest rises to a terminal tip (Crest-high decile pt=0.74) and pSom rises (rho +0.46). joint Gast&Crest enrichment 0.27 (<1 = inverse gradient, not a co-expression hub).

Gate 3 — cross-repression — NA

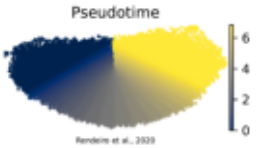
Not applicable — the manifold is non-branching (principal curve, 2 tips / 0 forks; a PPT tree throws 6-7 spurious forks on this near-round cloud). No bifurcation, so the branching-only hub cross-repression test is not run (correlation / modules notebooks are not produced, per the deliverable spec).

STEP FIGURES

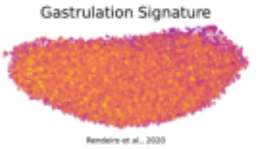
01 · Samples (Gate 1)



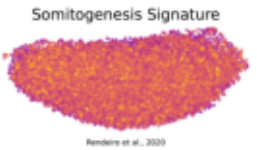
02 · Pseudotime



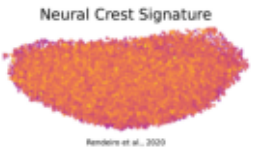
03 · Gastrulation



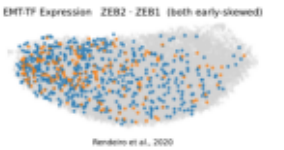
03 · Somitogenesis



03 · Neural Crest



04 · EMT-TFs



Dataset: CLL peripheral blood (Rendeiro 2020), 10x, 18,452 malignant cells over 4 patients (7 longitudinal samples); single diffuse-cycling body (~44% cycling, cell-cycle-free axis).

Verdict: Pattern reproduces in the non-branching form (Linear-pSom/Crest): one continuous gastrulation -> neural-crest axis, Crest terminal. G1 GO / G2 GO / G3 NA. CLL barely expresses EMT-TFs — only ZEB2 and ZEB1 clear the detection floor, both root-skewed (no terminus EMT-TF).

PARAMETER CHOICES

Sample threshold	geom-mean 284 (dominant cliff 1012->80, ratio 12.6) -> 7/11 samples; 18,452 cells after islands
Overdispersed genes	528 (scFates find_overdispersed)
Integration	Harmony on sample, theta=10 (mixing plateau; strongest integration for patient-private CLL batch)
Embedding	n_pcs=20, n_neighbors=30, UMAP spread=1.3 (10x precedent scale); 205 cells / 3 islands removed
Geometry	Non-branching -> scFates principal curve (ElPiGraph), Nodes=20, epg_lambda=0.3, epg_mu=100
Topology check	2 tips / 0 forks; fold self-approach 2.58 (>1 fold-free); PPT tree 6-7 spurious forks; corr(x_lin,t)=0.92
Root	tip 0 = Gast/pSom-high, Crest-low developmental origin (derived from signature order)
Module means (raw)	Gast -0.020 · pSom +0.008 · Crest +0.010 (Linear-pSom/Crest)
Cell cycle	43.8% cycling, diffuse (single lobe); axis cell-cycle-free (corr modules vs cycling ~0)
EMT-TFs shown	ZEB2 + ZEB1, both root-skewed (no terminus EMT-TF); >3% gate; FOXC/PRRX/SNAI/TWIST below floor